MECH 325 - Midterm Book 1
Closed Book

Thursday, October 24, 2024

Name: Solutions

Student Number: _____

Circle your section

Section 102 (11:00 Lecture)

Section 101 (9:30 Lecture)

Signature: _____

Part 1 MC Mark _____ / 12

Part 1 SA Mark _____ / 10

Part 2 LA Mark _____ / 38

Total _____ / 60

Instructions

There are two parts to this exam with different instructions for each. Please read carefully.

Part 1 – Closed-Book (22 marks)

Multiple choice (12 marks: 1 marks per question). Complete all 12 questions by marking your response on the paper.

Short answer (10 marks). Complete all 5 questions by marking your response in this exam booklet.

After you have completed Part 1, hand in your paper in order to begin the Open Book portion. You may not return to Part 1 after you hand it in.

Part 2 – Open-Book (38 marks)

Long answer (38 marks): Complete all parts to the questions for Part 2 by marking your responses on the exam paper. This portion of the exam is open-book and open-notes.

NO COMMUNICATION DEVICES!

Part 1A: Multiple Choice (12 Points)

Answer all 12 questions. Each is worth 1 mark. Choose the single best response – a given question may have more than one choice that is correct but marks will only be given for the *best* answer.

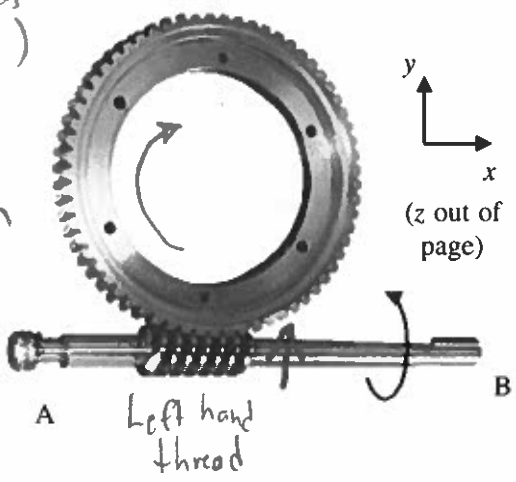
1. **What is the best description of the type of motion between the teeth of two properly meshed worm gears?**
   a) Purely rolling
   b) Purely sliding
   c) Some rolling when teeth engage and disengage, but predominantly sliding
   d) Some sliding when teeth engage and disengage, but predominantly rolling
   e) Approximately equal parts sliding and rolling

2. **For spur and helical gears, the pressure line is tangent to which circles?**
   a) Pitch Circle
   b) Base Circle
   c) Addendum Circle
   d) Dedendum Circle
   e) Clearance Circle

3. **The worm gear shown has a 1" pitch diameter and has two threads. The gear has an 8" pitch diameter and 96 teeth. If the worm turns 24 revolutions in the direction indicated, how far does the gear turn?**

   a) 8 rotations clockwise
   b) 4 rotation clockwise
   c) ½ rotation clockwise
   d) ½ rotation counter-clockwise
   e) 4 rotations counter-clockwise

Lead is 2 threads (pitch)
∴ 24 revs moves gear 48 teeth = ½ rotation



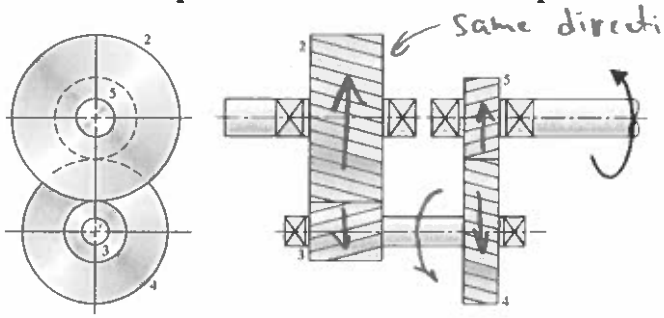
4. **Which of the following types of gears is appropriate for high-load transmission of motion between parallel shafts with no axial loading on either shaft?**
   a) Bevel
   b) Hypoid
   c) Worm
   d) Herringbone
   e) Helical

5. **A spur gear has 36 teeth, a 20° pressure angle, and a pitch diameter of 3.0 in. What is the diametrical pitch of the gear?**

$P_d = \frac{N}{d} = \frac{36}{3} = 12 \text{ teeth/inch}$

a) 4.0
b) 6.0
c) 9.0
d) 12.0
e) 20.0

6. **For the gear train shown, gear 5 is the driving pinion Gear 5 has 17 teeth, Gear 4 has 34 teeth. Gear 3 has 16 teeth and Gear 2 has 40 teeth. An 1800 rpm motor is driving the gear box in the Clockwise direction. What is the speed and direction of the output shaft?**



$e = \frac{17}{34} \cdot \frac{16}{40} = 0.2$

$n_f = e\, n_i = 0.2(1800)$

$n_f = 360 \text{ rpm}$

a) 360 rpm clockwise
b) 360 rpm counter clockwise
c) 720 rpm clockwise
d) 720 rpm counter clockwise
e) 300 rpm clockwise

7. **The motor attached to the gearset in Question 6 above is rated at 750 Watt. What is the torque on the output shaft?**

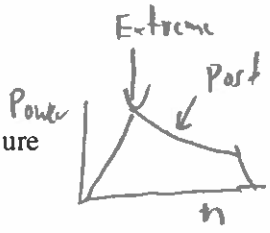
$P = T\omega$

$750 = T(360 \text{ rpm})(1 \text{ min}/60 \text{ sec})(2\pi \text{ rad}/1 \text{ rev})$

$\therefore T = 19.9 \text{ N-m}$

a) 15 N-m
b) 19.9 N-m
c) 23.9 N-m
d) 30 N-m
e) 37.7N-m

8. **The term *"Post-Extreme Horespower"* refers to the following:**



a) The breaking strength of a synchronous belt
b) The power rating for a chain where roller fatigue or excessive wear are the modes of failure
c) The power rating for a chain where link fatigue failure is the primary mode of failure
d) The maximum horsepower rating for a chain
e) The smallest degree of lubrication

9. **Which of the following components of belt tension present in V-Belts but normally ignored in flat belts that is the main consideration when calculating belt life?**

a) Initial tension (F_i)
b) Bending tension (F_b)

c) Centrifugal tension (F_c)
d) Tight-side tension (F_1)
e) Slack-side tension (F_2)

10. Which of the following statements regarding gears is TRUE?

a) Spur gears can take higher loads than helical gears
b) Helical gears can never be used for right angle drives (shafts 90°)
c) Lower pressure angles allow for higher transverse forces between gears
d) Pinions with only 10 teeth can be used with a rack
e) Miter gears are a form of helical gears

11. For roller chains the term *chordal speed variation* refers to:

a) the speed of engagement of a chain with a sprocket
b) the relationship between the chain speed and the sprocket speed
c) the change in chain speed as the friction in the chain changes
d) the change in chain speed due to the slight rise and fall of the chain as it engages with the sprocket
e) the change in chain speed due to elongation of the chain

12. When specifying the working load of wire rope, which of the statements is FALSE:

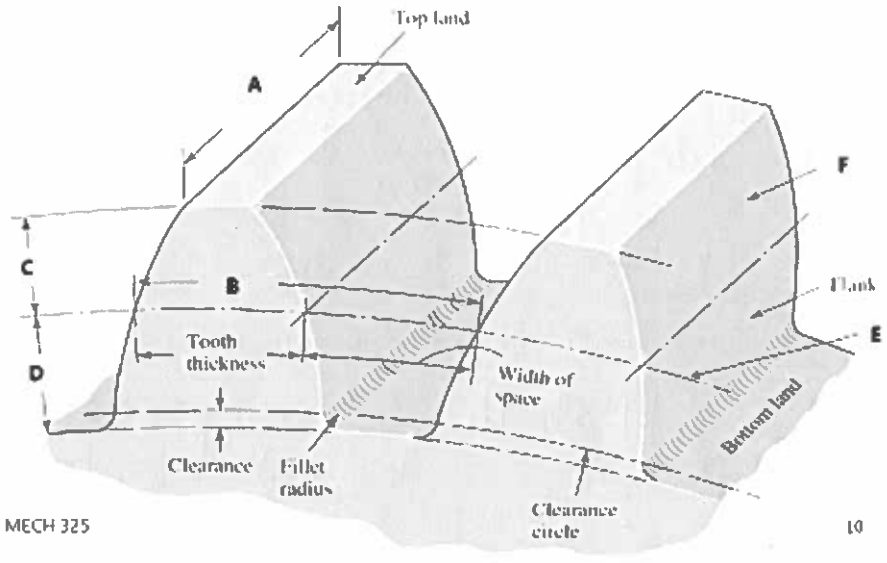
a. one must consider the modulus of elasticity of the wire rope
b. one must take in account the maximum allowable bearing pressure of the wire rope on the sheave
c. one must consider the application in order to specify a safety factor
d. one must determine the maximum sheave diameter
e. one must consider loads caused by sudden stops and starts

13. (Bonus) To convert from horsepower to ft-lbs/sec use the following conversion

a) Divide by 454
b) Divide by 33000
c) Multiply by 550
d) Divide by 550
e) Divide by 32.2

Part 1B – Short-Answer Questions (10 marks)

Answer all questions in the spaces provided.



14. Name the SIX missing terms (A through F) related to spur gears (3 marks)

A - Face Width
B - Pitch (Circular)
C - Addendum
D - Dedendum
E - Pitch Circle (Diameter)
F - Face

15. Which of the above terms are important for stress analysis of the gear teeth? Why (3 marks)?

The pitch, pitch circle, and width are key parameter for stress analysis.
Both bending & contact stress decrease as width increases.
As circular pitch increases, the tooth size increases [P_d lower] which reduces bending stress. Increasing pitch circle will reduce transmitted load, W_t, for the same torque, which reduces stress.

16. In gear design, what is meant by interference? How can it be avoided (2 marks)?

Interference relates to gear tooth contact that is no longer conjugate. It causes the teeth to bind and cause excessive wear.
Interference is avoided by having the teeth engage between the tangent points on the base circles formed by the pressure line. It is avoided by selecting pinions that have a suitable number of teeth for the application (typically 17-19 teeth)

17. What is meant by the "capstan equation?" How does it relate to the belt drives? (2 marks)

The capstan equation highlights the effectiveness of friction drives. It relates the ratio of taut and slack side tension with the friction coefficient and wrap angle.

$P_1/P_2 = e^{f\phi}$

P_1 = taut side tension
P_2 = slack side tension
f = coeff. of friction
ϕ = wrap angle

As either f or ϕ ↑ the ratio P_1/P_2 ↑

Note: "Rule of thumb" For every wrap of rope around the capstan, the tension drops by ½.

MECH 325 - Midterm Book 2
Open Book

Thursday, October 24, 2024

Name: Solutions

Student Number: _____

Circle your section

Section 102 (11:00 Lecture)

Section 101 (9:30 Lecture)

I adhered to the policy of No Communication Devices.

Signature: _____

Instructions

Please read carefully.

No Communication Devices Allowed!

There are **Two** Problems to complete.

Part 2 – Open-Book (38 pts)

Long answer: Complete all parts to the question by marking your response in the booklet provided. This portion of the exam is open-book and open-notes. **At the end of the exam, return this handout with your answers in the booklet.**

This is a multi-part problem. For each part, use the parameter values given. If you cannot complete one part, you may assume some values and move on in order to receive marks for subsequent parts. Clearly indicate your final answer to each part by drawing a box around it. Separate different parts with a line drawn across the page.

Long Answer Problem (38 marks)

Problem 1 (16 Marks)

A chain drive system is required to attach an electric motor to an industrial agitator. The following information relates to the equipment

Driver	AC Motor (High Torque)
Driven Machine	Hammer Mill
Service	8 hours/day
Input Speed	1150 rpm
Input Power	10 HP
Output Speed (Nominal)	450 rpm
Centerline Distance (Nominal)	36 inches
Design Factor	1.0

2 a) Specify a suitable service factor
5 b) Specify a chain size, and the number of teeth for the sprockets. Select the smaller sprocket to have less than 20 teeth. What is the final velocity ratio?
3 c) Specify the number of chain pitches (links) for the nominal centerline distance
2 d) Specify the type of lubrication regime required for this operation.
4 e) What is the safety factor for your design? Comment on the result

a) Using Table 7-17 (Moth) for electric motor and heavy shock (hammer mill)
$K_o = 1.5$ (2 mk)

b) Input speed = 1150 rpm Output Speed = 450 rpm
Desired velocity ratio = 1150/450 = 2.56 (1mk)

Options for Chain Drives. It is recommended to have the minimum sprocket size at 17 teeth

(i) 17 tooth sprocket → 43 tooth (Driven) ratio = 2.53
(ii) 18 " " → 46 " " ratio = 2.56
(iii) 19 " " → 49 " " ratio = 2.58

We have a perfect match with 18 tooth sprocket (DriveR) (1mk)

$VR_{final} = 46/18 = 2.56$ DriveR = 18 teeth
DriveN = 46 teeth

Chain size

(a) M of H

Design H.P = $K_s H_{nom}$ = 1.5 (10 HP) = 15 HP

a) #40 chain Table 7-14 H_{tab} @ 1000 rpm = 7.69 HP } note: pre-extreme
H_{tab} @ 1200 rpm = 9.17 HP }

1 mk
H_{tab}

H_{tab} @ 1150 = 8.8 HP

Need at least 3 strand ∴ too light
reject

b) #60 chain Table 7-15 H_{tab} @ 1000 rpm = 25.73 HP } Post-extreme
H_{tab} @ 1200 rpm = 24.81 HP }

1 mk

∴ H_{tab} @ 1150 rpm = 25.04 HP

Select Single Strand #60 chain Pitch = 0.75 in

(ii) Shigley Design HP = 1.5(10) = 15 HP

a) #50 chain Table 17-20 H_{tab} @ 1000 rpm = 10.7 HP }
H_{tab} @ 1200 rpm = 12.6 HP } ↑
H_{tab} @ 1150 rpm = 12.1 HP pre-extrem

K_1 = 1.06* Table 17-22 for 18 teeth

* pre-extreme as H_{Tab} increase to 14.4 HP @ 1400 rpm

K_2 = 1.7 [two strands]

$H_A = K_1 K_2 H_{Tab}$ = (1.06)(1.7)(12.1) = 21.8 HP

Select Duo Strand #50 Chain Pitch = 0.625 inch

b) #60 chain Table 17-20 H_{tab} @ 1000 rpm = 18.3 HP } pre-extrem
H_{tab} @ 1200 rpm = 21.6 HP }

K_1 = 1.06* for 18 teeth
* pre-extreme

H_{tab} @ 1150 rpm = 20.8 HP

K_2 = 1.0 [one strand]

H_A = (1.06)(1.0)(20.8) = 22.0 H

Select Single Strand #60 Chain, Pitch = 0.75 inch

c) Nominal Centerline Distance $C = 36$ inches

(1mk) $L/p = \frac{2C}{p} + \frac{N_1+N_2}{2} + \frac{(N_2-N_1)^2}{4\pi^2 C/p}$ Mott Eq'n 7-18 (C = # links) Shigley Eq'n 17-34 (C = inches)

i) For #60 chain $P = 0.75''$ $N_1 = 18$ $N_2 = 46$

$L/p = \frac{2(36)}{0.75} + \frac{18+46}{2} + \frac{(46-18)^2}{4\pi^2 36/0.75} = 128.4$ pitches (2mk)

Note for Mott 7-18 you need to specify CD as a distance in links! CD = 36"/0.75 = 48 links!

Round up to 129 links so that odd # links mesh with even number of teeth on sprocket

Links = 129 @ #60 chain

Shigley ii) For #50 chain $P = 0.625''$ $N_1 = 18$ $N_2 = 46$

$L/p = \frac{2(36)}{0.625} + \frac{18+46}{2} + \frac{(46-18)^2}{4\pi^2 36/0.625} = 147.5$ pitches

Round down to 147 links for odd number on even sprocket

Links = 147 @ #50 chain

(2mk) d) Lubrication Regime

Mott Table 7-15 Type B
Shigley Table 17-20 Type B
} Bath or Disk Lubrication

e) Safety Factor: $n_{SF} = \frac{H_{allowable}}{H_{design}} = \frac{H_{all}}{H_d}$

1) Mott Table 7-15 $H_{tab} = 25.04$ HP (see part b))

(2mk) $n_{SF} = 25.04/15 = 1.67$ #60 Chain Single Strand

2) Shigley #50 chain Table 17-20 $H_{tab} = 21.8$ (part b)

$n_{SF} = 21.8/15 = 1.45$ #50 Chain Duo Strand

3) Shigley #60 chain Table 17-20 $H_{tab} = 22.0$ (part b)

(2mk) $n_{SF} = 22.0/15 = 1.47$ #60 Chain Single Strand

(2mk) All safety factors are >1.0 so our design is considered safe

Note Mott S.F & Shigley S.F. for #60 chain are not equal as the chain

Problem 2 (22 Points)

Part 1 of 3

(8 points) Instead of using a chain drive, it is proposed that a gear box be used to provide the required **nominal** speed of 450 rpm. In this new design the power is increased to 20HP. Do an analysis on the following full depth involute **HELICAL** gears:

Note: you can choose imperial or metric units

Driver	AC Motor (Normal Torque)
Driven Machine	Hammer Mill
Service	8 hours/day
Input Speed	1750 rpm
Output Speed	450 rpm (nominal)
Input Power	20 HP (15 kW)
Pressure Angle (Normal)	20°
Helix Angle	20°
Diametrical Pitch (Normal)	10 teeth/inch (module = 2.5 mm.)
Pinion	19 teeth
Gear	? teeth

Determine the following:

1 a) Given 19 teeth for the pinion, select a gear and determine the gear ratio (velocity ratio)
2 b) Determine the pitch diameter for pinion and gear. What is the Diametrical Pitch (Tangential)?
3 c) Determine the pitch line speed
4 d) Determine the tangential, radial, and axial forces for the gears

a) Velocity Ratio $VR = 1750/450 = 3.889$

Number of teeth for gear $= (3.88)(19) = 73.89$

Select 74 tooth gear

Final $VR = 74/19 = 3.895$ Output Speed $= 449.3$ rpm

<1% difference

1 mk

b) Diametrical Pitch $P_d = P_{nd} \cos\psi = 10 \cos 20°$ [Moh 8-16, Shigley 13-18]

(trans) (nor)

Tangential $P_d = 9.397$ teeth/inch

1 mk

$d_p = N/P_d$

Pinion pitch dia $= 19/9.397 =$ 2.022 inches $= d_{pinion}$

Gear pitch dia $= 74/9.397 =$ 7.875 inches $= d_{gear}$

1 mk

c) Pitch Line Speed = ωr

1 mk

$$V_t = \frac{\pi d n}{12} = \frac{\pi (2.022)(1750 \text{ rpm})}{12} \quad \text{[Mott 9-1]}$$

$$\boxed{V_t = 926.4 \text{ fpm (ft/min)}}$$

d) Tangential, Radial, & Axial Forces

i) Tangential Force, W_t $\quad W_t = T/(d_p/2)$

$$\text{Power} = W_t \cdot V_t \rightarrow \text{Power (HP)} = \frac{W_t(\text{lbf})\, V_t(\text{ft/min})}{33000} \quad \text{[Mott 9-9]}$$

1 mk

$$W_t = (20 \text{ HP})(33000)/926.4 \text{ fpm}$$

$$\boxed{W_t = 712.4 \text{ lbs}}$$

ii) Radial Force, W_r

$$W_r = W_t \tan \phi_t \quad \text{[Mott 10-8, Shigley 13-46]}$$

1 mk

$$\tan \phi_t = \frac{\tan \phi_n}{\cos \psi} \quad \text{[Mott 10-1, Shigley 13-19]}$$

$$\phi_t = \tan^{-1}\left[\frac{\tan \phi_n}{\cos \psi}\right] = \tan^{-1}\left[\frac{\tan 20^\circ}{\cos 20^\circ}\right] = 21.17^\circ$$

1 mk

$$W_r = (712.4)(\tan 21.17) \quad \boxed{W_r = 275.9 \text{ lbs}}$$

iii) Axial Force, W_x

$$W_x = W_t \tan \psi = (712.4)(\tan 20^\circ) \quad \text{(Mott 10-9 / Shigley 13-40)}$$

$$\boxed{W_x = 259.3 \text{ lbs}}$$

1 mk

Note: Normal Force $W = W_t/(\cos \phi_n \cos \psi) = 712.4/(\cos 20^\circ \cos 20^\circ)$

$$W = 806.8 \text{ lbs}$$

Note: axial pitch $p_x = p_t/\tan \psi \quad \therefore p_x = 0.334/\tan 20^\circ = 0.917''$

p_t = circular pitch $= \pi/p_t = \pi/9.397$

$p_t = 0.334$

Part 2 of 3

(10 Points) Determine the AGMA **bending** stress number for the 19 tooth **pinion ONLY.**

Given:

Face Width	1.5 inch (40 mm.)
Pressure Angle	20°
Gear Quality	$A_v = 8$ ($Q = 9$)
Material for Pinion & Gear	Steel
Gear Box Enclosure	Commercial Quality

✓	S Geometry factor	J = ?
	Overload Factor	**$K_o = 2.25$** (uniform power source and heavy shock)
✓	Dynamic factor	K_v = ?
✓	Load Distribution factor	K_m = ? (For Shigley Text: $C_{mc} = C_{pm} = C_e = 1.0$)
✓	Size Factor	K_s = ?
✓	Rim thickness factor	K_B = ?
✓	Bending stress number	s_t = ?

For bending stress number [Mott 9-23, Shigley 14-15]

$$s_t = \frac{W_t P_d}{F J} K_o K_s K_m K_B K_v$$

i) Geometry Factor J

For Helical Gears use Fig 10-6 (M.H) or Fig 14-7 (Shigley) (2mk)

(Mott) From Fig 10-6 $J = 0.46$ (20 teeth @ $\psi = 20°$)

Modifying Factor $K_j = 1.0$ (mesh with 74 teeth ~ 75 teeth)

$\therefore J = 0.46$ (M.H)

(Shigley) From Fig 14-7 $J = 0.50$ (20 teeth @ $\psi = 20°$)

Fig 14-8 $K_j = 1.0$ (mesh with 74 teeth ~ 75 teeth)

$\therefore J = 0.50$ (Shigley)

ii) $K_o = 2.25$ Given

iii) Dynamic Factor, K_v

$V_t = 926.4$ fpm, $A_v = 8$ ($Q = 9$) gear quality

For Helical Gears, we can use the Spur Gear dynamic factor

From MoTT 9-16

$K_v = 1.18$ (using Figure) (2mk)

$K_v = 1.17 - 1.19$

$$K_v = \left[\frac{C}{C + \sqrt{V_t}}\right]^{-B}$$

$B = 0.25(A_v - 5)^{0.667} = 0.52$

$C = 50 + 56(1.0 - B) = 76.87$

$$K_v = \left[\frac{76.87}{76.87 + \sqrt{926}}\right]^{-0.52}$$

$K_v = 1.19$

iv) Load Distribution Factor, K_m F = Face Width = 1.5 inches

$K_m = 1 + C_{pf} + C_{ma}$ (M+H 9-17)

C_{pf} = pinion proportion factor

Use M+H Fig 9-12 $F/D_p = 1.5/2.02 = 0.74$

From Fig 9-12 $C_{pf} \simeq 0.04$ @ F = 1.5 & $F/D_p = 0.74$

$$C_{pf} = \frac{F}{10 D_p} - 0.0375 + 0.0125 F = \frac{1.5}{10(2.02)} - 0.0375 + 0.0125(1.5)$$

$C_{pf} = 0.055 \sim 0.06$ (2 mk)

C_{ma} = mesh alignment factor

Assume Commercial enclosed gear unit

From Fig 9-13 $C_{ma} \simeq 0.15$ @ F = 1.5

$C_{ma} = 0.127 + 0.0158F - 1.093 \times 10^{-4} F^2 = 0.127 +$

$C_{ma} = 0.150$

$K_m = 1.0 + 0.06 + 0.15 = \underline{1.21}$ (upper end of range)

v) Size Factor K_s (1 mk)

$K_s = 1.0$ since $P_D = 10 > 5$

vi) Rim Thickness Factor K_B (1 mk)

$\underline{K_B = 1.0}$ Assume pinion gear is solid since $d_p \sim 2$ Dinches

$$s_t = \frac{W_t P_d}{F J} K_o K_s K_m K_B K_v = \frac{(712.4 \text{ lb})(9.397 \frac{\text{teeth}}{\text{in}})}{(1.5)(0.46)} (2.25)(1)(1.21)(1)(1.1^{\circ}$$

$\boxed{s_t = 31432 \text{ psi} = 31.4 \text{ ksi}}$ (M+H value for J) (2 mk)

$\boxed{\sigma_b = s_t = 28918 \text{ psi} = 28.9 \text{ ksi}}$ (Shigley value for J

Note: Shigley & M+H use differing teeth profiles. Hence the differing values for J

Part 3 of 3

(4 points) Select a suitable through hardened steel for the **pinion**. Specify the alloy and heat treatment.

The gear is expected to operate for the machine 200 days/year for 5 years. You can also assume the pinion and gear have the same Brinell hardness. State all assumptions.

What is your safety factor for bending stress? Comment on the result

Would you expect the contact stress safety factor to be different for this gear? Explain.

We require an acceptable material strength number that gives a safety factor, S.F.) of approx $1.0 < SF < 1.5$ as per AGMA

$$S_t < S_{at}' = S_{at} \frac{Y_N}{(S_F)(K_R)(K_T)} \qquad \text{Moff 9-25}$$

$$\sigma_b < \sigma_{b,all} = \frac{S_t}{S.F} \frac{Y_N}{K_T K_R} \qquad \text{Shigley 14-17}$$

Y_n = Stress Cycle Factor with N = # of cycles

$N = 8\ hr/day \cdot 200\ days/yr \cdot 5\ years \cdot (60\ min/hr) \cdot 1750\ rph$ (part 1)

$N = 8.4 \times 10^8$ cycle

1 mk

From Shigley Fig 14-14 & Moff Fig 9-21

$Y_N \sim 0.95 @ 8.4 \times 10^8$; $Y_n = 1.3558 N^{-0.0178} = 0.94$

Assume $K_R = 1.0$ for 99% reliability is acceptable

Assume $K_T = 1.0$ at $T < 120°$ for this application

1 mk

$$\min S_{at}' = S_t \frac{K_T K_R SF}{Y} = \frac{31.4(1)(1)(SF=1)}{0.94} = 33.4\ ksi\ [Moff]$$

$$\min \sigma_b = 28.9\ ksi \frac{(1)(1)(1)}{0.94} = 30.8\ ksi\ [Shigley]$$

For through-hardened Grade 1 gears use Fig 9-18 (Moff) Fig 14-2 (Shigley)

1 mk

Required $HB = (S_{at} - 12.8)/0.0773$

with $S_{at} = 33.4$ ksi $HB_{min} = 267$ [Moff]

with $\sigma_b = 30.8$ $HB_{min} = 233$ [Shigley]

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Material Selection:

Several alloys are available for use in this application

Let's select 4140 Chro-Moly alloy that is recommended for gears [See Table 9-8]

(i) From Appendix 3 [Moth] with $s_t = 33.4$ ksi $HB_{min} = 267$

Select 4140 OQT 1000 with HB = 341 & elongation = 17%

With HB = 341 bending strength number, s_{at} is:

$$s_{at} = 0.0773\,HB + 12.8\ \text{ksi} = 0.0773(341) + 12.8 = 39.15\ \text{ksi}$$

Safety Factor $S.F = \frac{S_{at} Y}{K_T K_R s_t} = \frac{39.15(0.95)}{33.4} = 1.11 = S.F$

(ii) From Shigley Table A-21 and $\sigma_b = 30.8$ ksi HB = 259 min

Select 4140 QET (1000) with HB = 285

$$s_{at} = 0.0773(285) + 12.8 = 34.8\ \text{ksi} \qquad \sigma_{b,all} = \frac{s_{at} Y_N}{K_T K_R}$$

Imk

$$S.F = n_b = \frac{\sigma_{b,all}}{\sigma_b} = \frac{34.8(0.95)}{30.8} = 1.07 = S.F$$

Safety factor is acceptable according to AGMA standards as it falls between 1.0 and 1.5

Our experience has shown that our contact stress safety is normally lower than that for bending. It could fail in contact stress but we would need to conduct a full investigation using the AGMA contact stress equations. We may need to select a stronger alloy or a gear that has been carburized.

Also we should check that the face width and axial pitch meet the recommended ratio of $F/p_x > 2.0$

From part 1 we determined $p_x = 0.917$ inches $\therefore \frac{1.5}{0.917} < 2.0$ ∴ select wider gear say 1.5"